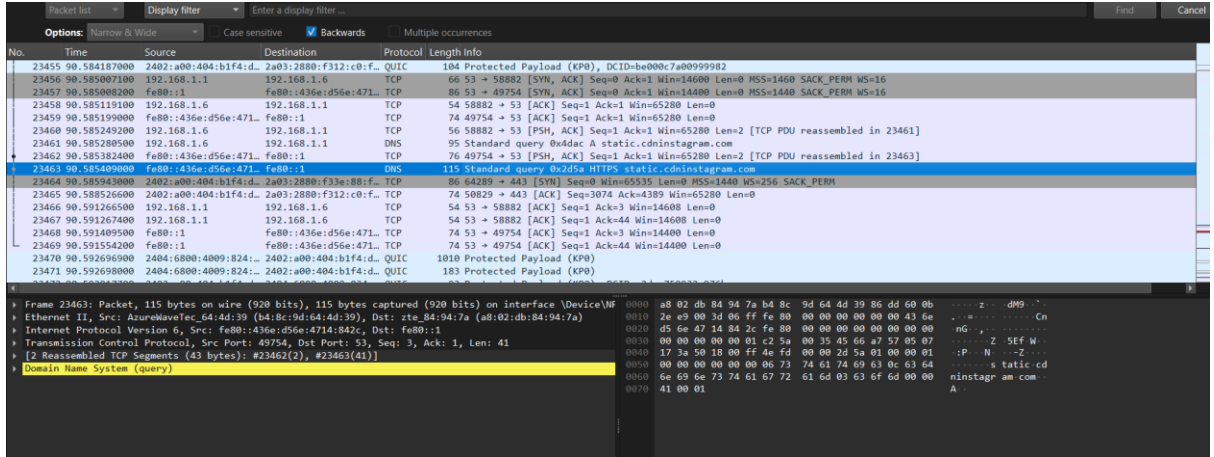


Day 1 Introduction to Networking & osi model

1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.



Wireshark Capture – Protocols Observed

I installed **Wireshark** on my laptop and started capturing packets on my main network interface. I then visited a website and stopped the capture after some traffic was generated.

The following protocols were observed in the packet list:

Protocol	Purpose
DNS	Resolves domain names such as static.cdninstagram.com into IP addresses.
TCP	Provides reliable communication between devices.
QUIC	A modern transport protocol used by applications such as HTTPS/HTTP/3.

Observation: The capture shows DNS queries for the website and TCP/QUIC traffic carrying the actual network communication.

2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

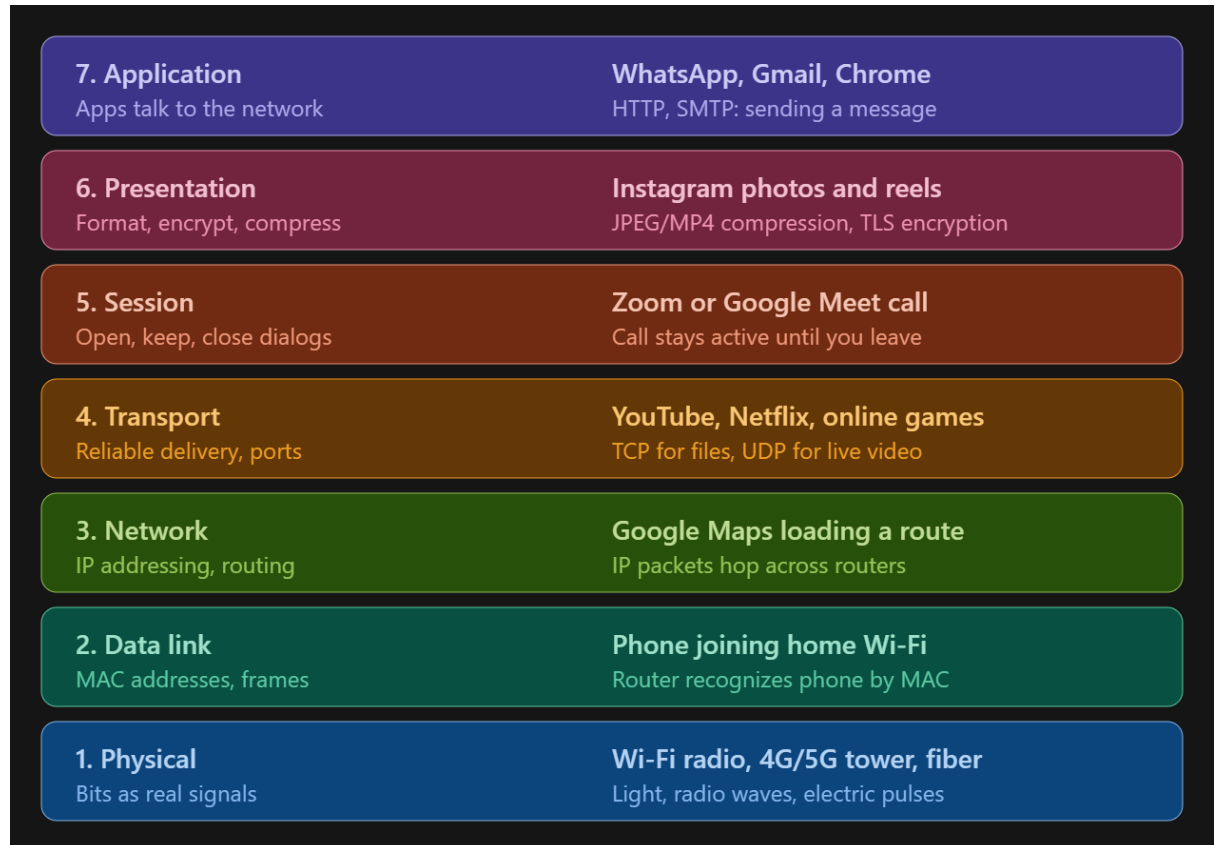
Wireshark Packet Analysis

I selected three packets from my Wireshark capture and mapped them to the OSI model based on the **Protocol** and **Info** columns.

Packet	Protocol	OSI Layer	Explanation
23463	DNS	Application Layer (Layer 7)	The packet contains a DNS query for static.cdninstagram.com, which is used to resolve a domain name to an IP address.
23464	TCP	Transport Layer (Layer 4)	The packet shows a TCP connection to port 443 , used for reliable communication.
23455	QUIC	Transport Layer (Layer 4)	QUIC provides transport-level communication and is commonly used with HTTP/3.

Conclusion: The capture contained protocols operating at different OSI layers. DNS mainly represents the **Application layer**, while TCP and QUIC mainly represent the **Transport layer**.

3. Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).



4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

When I send a message on WhatsApp, the message passes through different layers of the OSI Model. In the **Application Layer (Layer 7)**, I type and send the message using WhatsApp, and the application prepares it for transmission. In the **Presentation and Session Layers (Layers 6 and 5)**, the message is formatted, encrypted, and the communication session between devices is maintained.

In the **Transport Layer (Layer 4)**, the message data is divided into smaller parts and transport protocols help deliver it reliably. At the **Network Layer (Layer 3)**, IP addresses are used to determine the route for the message to reach the destination. The data is then passed through different networks and routers toward the receiver. At the **Data Link Layer (Layer 2)**, MAC addresses are used to transfer the data between devices on the local network, such as my phone and Wi-Fi router. Finally, at the **Physical Layer (Layer 1)**, the data is converted into radio, electrical, or other signals and transmitted through Wi-Fi, mobile networks, or cables. On the receiver's device, the process happens in reverse until the message appears in WhatsApp.